

Full Length Article

Comparative analysis of machine learning models for predicting dielectric properties in MoS₂ nanofiller-reinforced epoxy composites

Atul D Watpade^a, Sanketsinh Thakor^b, Prince Jain^{c,*}, Prajna P. Mohapatra^d, Chandan R. Vaja^e, Anand Joshi^c, Dimple V. Shah^a, Mohammad Tariqul Islam^{f,*}

^a Department of Physics, Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat 395007, India

^b Department of Applied Science and Humanities, Parul Institute of Technology, Parul University, Vadodara, Gujarat 391760, India

^c Department of Mechatronics Engineering, Parul Institute of Technology, Vadodara, Gujarat 391760, India

^d Department of Physics, Indian Institute of Technology Guwahati, Assam 781039, India

^e Department of Physics, Electronics and Space-Science, Gujarat University, Ahmedabad, Gujarat 380009, India

^f Department of Electrical, Electronic and Systems Engineering, Faculty of Engineering and Built Environment, Bangi, Selangor 43600 UKM, Malaysia

ARTICLE INFO

Keywords:

Epoxy composites
Dielectric properties
Machine learning
MoS₂
Nanofillers

ABSTRACT

This research investigates the dielectric properties of nano epoxy composites by incorporating various concentrations of MoS₂ into epoxy resin. The study explores the impact of synthesized nanoparticles on undoped epoxy composites, specifically focusing on their potential applications in dielectric materials. The experimental synthesis and characterization of nanoe epoxy composites typically involve time-consuming and expensive methods. This study compares five machine learning (ML) models—random forests, decision trees, extra trees, XGBoost, and gradient boosting—in order to predict the frequency-dependent dielectric constants in these composites under different nanofiller variations in order to address this challenge. To ensure robust model performance, training is carried out on different subsets of the dataset, ranging from 60% to 30%, while the remaining portions are reserved for testing purposes (40% to 70%). The main objective of the study is to assess the performance of each regressor technique using metrics such as adjusted R² score, MSE, RMSE, and MAE, in which the ET regressor excels. The ET method demonstrates exceptional performance, achieving an adjusted R² value of 0.9977 and 0.9912 for target variables ϵ' and ϵ'' , respectively when tested with a size of 0.4. The findings underscore the potential of ML models for precise and efficient prediction of frequency-dependent dielectric constants of nanoe epoxy composites with various concentrations of nanofillers, offering an alternative to time-consuming and expensive laboratory work.

1. Introduction

The investigation of dielectric characteristics in polymer composites doped with MoS₂ nanoparticles has gained significant growth in the field of nanopolymer materials in recent times. Specifically, epoxy resin is widely used as a base matrix because of its exceptional mechanical, electrical, and thermal characteristics as well as its affordability and structural strength [1,2]. The use of nano-fillers such as MoS₂ for epoxy reinforcement appears promising, and the dielectric properties of MoS₂ nanoparticle-doped nano-epoxy composites can be precisely tailored to improve capacitor, insulator, and energy storage system performance

[3]. Nagachandrika et al. [4] found that the dielectric constant of MoS₂-filled epoxy composites decreased as the supply voltage frequency increased. As the dielectric constant of epoxy composites increased above 0.5 wt% MoS₂, there appeared to be a correlation between the concentration of MoS₂ and their dielectric behaviour. Zhen Liu et al. investigated the thermal conductivity of epoxy composites enhanced with microspheric molybdenum disulfide (S-MoS₂). They found that using the appropriate microspheric molybdenum disulfide increased the dielectric constant and decreased dielectric loss [5]. The unique properties and broad range of applications of these composites enriched with MoS₂ nanoparticles have sparked interest in research, especially in

Abbreviations: RMSE, Root Mean Squared Error; MSE, Mean Squared Error; MAE, Mean Average Error; ET, Extra Tree; ML, Machine Learning; XGBoost, Extreme Gradient Boosting.

* Corresponding authors.

E-mail addresses: princece48@gmail.com (P. Jain), tariqul@ukm.edu.my (M. Tariqul Islam).

<https://doi.org/10.1016/j.asej.2024.102754>

Received 23 October 2023; Received in revised form 1 February 2024; Accepted 29 February 2024

Available online 14 March 2024

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electronics [6]. The integration of MoS₂ nanoparticles and nano epoxy resin has accelerated electronic component innovation and increased their utility in critical applications like electrical insulation and short-circuit protection, ensuring the reliable operation of various electronic systems [7–9]. However, current scientific literature does not provide detailed research on the dielectric properties of epoxy nanocomposites doped with MoS₂ nano-fillers, particularly using a machine learning approach tailored to this system. Additionally, material preparation is frequently a lengthy, laborious, and time-consuming process [10–13]. Many samples are required to create a composite material with the desired properties, which increases the cost and time. G. Pilania addresses these issues through quasi-experimental data and powerful prediction algorithms [14]. The objective of this method is to overcome the drawbacks of conventional experimental methods by creating mathematical models and algorithms that can make accurate predictions for hypothetical samples. When dealing with large datasets, traditional mathematical analyses can be ineffective; however, machine learning (ML) approaches show much greater competence in assessing and forecasting material characteristics [15–19].

Machine learning techniques have quickly gained popularity in materials science due to their exceptional predictive capacity for complex data [20]. Several regression algorithms have been used to predict material properties over the last decade, including random forest (RF), decision trees (DT), extra trees (ET), Extreme Gradient Boosting (XGBoost), Gradient Boosting Machine (GBM), support vector machines (SVM), and k-Nearest Neighbour (KNN) [21]. Taser et al. examined the impact of various doping concentrations on the dielectric properties of phthalocyanine (Pc)-doped nematic liquid crystal (NLC) composites, employing traditional regression algorithms including XGBoost, RF, and DT Regression [22]. A decision tree comprises decision nodes, which segment data, and leaf nodes that determine the target value. DT regression uses this structure to predict numeric outcomes through iterative data [23,24]. The Random Forest (RF) algorithm employs the technique of bagging to generate a collection of decision trees. RF regression combines multiple decision trees that are trained on different subsets of data [25]. The final prediction of RF regression is obtained by averaging the outcomes of individual regression trees. The quantification of the tree population within a forest is an essential parameter for RF analysis [26]. As the RF algorithm, Extreme Gradient Boosting (XGBoost) employs a large number of decision trees. The main difference between RF and XGBoost is that RF constructs its regression trees in isolation, whereas XGBoost constructs its trees in a predetermined order. [16,17,27,28].

Notably, Golin and Stefani employed SVM to forecast material properties for organic LEDs and LCD applications [29]. Sigaki et al. utilized the KNN technique to estimate key parameters of LCs from

texture images [30], while Inokuchi et al. utilized various ML algorithms to make predictions regarding phase transition temperature of binary liquid crystal (LC) molecules [31]. The utilization of ML techniques such as Random Forest and Gaussian process techniques for predicting the dielectric properties of materials has been explored with promising results [11]. Jain et al. investigated the predictive capabilities of the extra tree regression method for forecasting the absorption of metamaterial absorbers [32,33]. The ETR algorithm operates on the same principles as the RF algorithm [34]. To avoid overfitting, ETR trains each base learner with a random subset of features. While RF trains regression trees using the bootstrap method, ETR trains each tree using the entire training set. X. Lin et al. utilized the GBR model to accurately predict the electronic contribution of the polycrystalline dielectric constant, based on structural and elemental information [35]. Gradient boosting regression tree algorithms leverage ensemble learning, combining individual decision trees as weak learners to form robust forecasting models. This iterative approach effectively reduces forecast errors, making it adaptable to non-linear relationships and versatile in handling differentiable loss functions [36]. Parikh et al. demonstrated machine learning models, including linear regression (LR), DT regression, gaussian process regression (GPR), and GBR, to accurately predict low-frequency responses in metal halide perovskite solar cells [18,37,38].

This research focuses on evaluating machine learning models for precise and efficient prediction of frequency-dependent dielectric constants in nanoe epoxy composites at varying weight percentages. The objective is to eliminate the need for laborious and expensive experimental synthesis and characterization methods. This is accomplished by enhancing machine learning models to provide accurate predictions for different weight percentages and optimizing them across a broad frequency range. A hydrothermal method was employed to synthesize MoS₂ nanoparticles, and their impact on the complex dielectric permittivity of neat epoxy was investigated. The study presents complex dielectric permittivity in terms of dielectric constant (ϵ') and dielectric loss (ϵ''), with a comprehensive comparative analysis of five distinct regression models, including DT, RF, ET, XGBoost, and GBR. These findings underscore the transformative potential of ML models in advancing the comprehension and application of composite materials in real-world scenarios.

2. Experimental: Preparation of the composites

To synthesize the MoS₂ nanoparticle, homogeneous solution, 5.16 g of ammonium heptamolybdate ((NH₄)₆Mo₇O₂₄) and 9.50 g of thiourea (CH₄N₂S) were dissolved in 150 ml of ionized water. After 20 min of mixing, the mixture was placed into a 450 ml Teflon-lined hardened steel autoclave. It was then securely fastened and subjected to

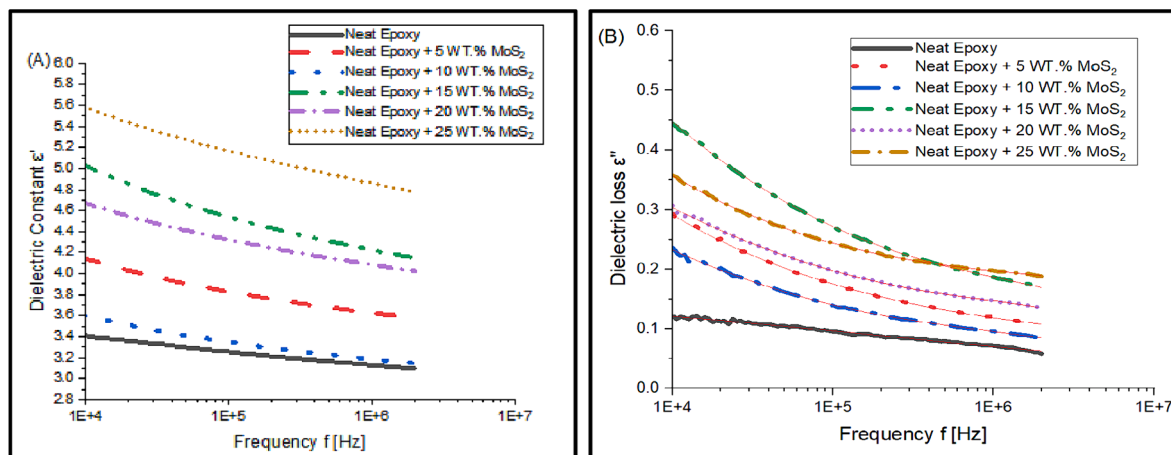


Fig. 1. (A) Dielectric constant (ϵ') (B) dielectric loss (ϵ'') with respect to frequency of MoS₂ nanofillers doped nano-epoxy composite.

hydrothermal treatment in an electric stove at 220 °C for 24 h, with continuous stirring at 200 rpm. The autoclave achieved a temperature of 220 °C after 2.5 h of heating. Once the autoclave had naturally cooled, the resulting black precipitate was collected by filtration. The collected material underwent multiple washing cycles using high-purity ethanol and ionized water to remove impurities. Subsequently, the material was dried at 60 °C under vacuum conditions for 24 h using a hoover.

To create a 3-d crosslinked structure, commercial epoxy resin Bisphenol A-epichlorohydrin and hardener N(3-dimethylaminopropyl)-1,3-propylenediamine (Huntsman International, India) were used. The epoxy and hardener were thoroughly mixed in a weight percentage ratio of 100:80 (wt%) to create the neat epoxy sample. Likewise, to make nano-epoxy composites (MOS₂ Nanoparticle + Neat epoxy, MOS₂ nanofillers were dispersed in neat epoxy at concentrations of 1, 5, 10, 15, and 20 wt% using ultrasonication (45 min). Subsequently, the composite mixture comprising epoxy and MOS₂ nanoparticles was carefully introduced into plastic moulds with a disc-shaped configuration that had previously been treated with a layer of wax. The specimens had a diameter of 2.5 cm and a thickness of 4 mm. The samples that were prepared underwent vacuum desiccation prior to their utilisation in the experiment.

2.1. Dielectric Characterization

The MoS₂ nanoparticle reinforced epoxy composites' dielectric spectroscopy was measured using an Agilent E4980A LCR metre equipped with a 16451B-test fixture. At room temperature, the measurements were carried out in the frequency range of 10 KHz to 2 MHz. The real component (ϵ') was calculated for all samples using the equation below

$$\epsilon^*(f) = \epsilon' - j\epsilon'' = \alpha \left(\frac{C_p}{C_0} - j \frac{1}{\omega C_0 R_p} \right) \quad (1)$$

Capacitance, denoted by C_p , is measured with the epoxy composite sample sandwiched between the two test fixture plates, C_0 represents the capacitance in the absence of a sample, with the plates being equidistant or having equal thickness, R_p stands for parallel $R_{\text{equivalent}}$ with a sample of epoxy composites. and α is the correction coefficient.

Fig. 1(A) illustrates the relationship between the relative permittivity (ϵ') and the frequency for epoxy composites doped with MoS₂ nanoparticles. The dielectric constant of the pure epoxy resin and all composites of epoxy doped with MoS₂ nanoparticles exhibited a slight reduction as the frequency increased. The observed trend may be attributed to a reduction in orientation polarisation within the pure epoxy resin. The decrease in the relative permittivity value can be attributed to the ineffective alignment of functional groups at higher frequencies [39,40]. As shown in Fig. 1(A), synthesized nanoparticles-doped epoxy composites had a greater relative permittivity than the neat epoxy (epoxy + hardener without doping of MoS₂ nanofillers) composite. The addition of MoS₂ nanoparticles improves the composite's overall polarisation and electrical conductivity, increasing the dielectric constant. These findings support the use of MoS₂ nanoparticles in neat epoxy resin, which binds to itself to limit polar molecule mobility.

The nonlinear behaviour of nanoparticle-filled composites' dielectric constant can be attributed to the concentration of nanoparticle doping caused by a variety of factors [2]. Interfacial polarisation can be caused at the interfaces where epoxy resin and nanoparticles meet by the presence of nanoparticles in the resin. Furthermore, when nanoparticle concentrations are low, the dispersion of particles within the resin matrix is such that they are widely separated, thereby reducing the occurrence of particle-particle interactions. However, clusters of nanoparticles form as the nanoparticle concentration rises, influencing the dielectric properties through their interactions with each other. These agglomerates create regions with increased dielectric characteristics by

Table 1

Fitting parameters for both pure epoxy and nanoparticle-doped epoxy composites.

	wt. %	α	β	τ (s)	$\Delta\epsilon$	Statistics	
						Chi-Sqr Adj.	R-Square
MoS ₂ nanoparticles-doped epoxy composites							
	0	0.17	1	0.15352	3.27	9.50 x10 ⁻⁰⁷	0.999
	1	0.03	1	1 x10 ⁻⁰⁷	6.22	2.07 x10 ⁻⁰⁵	0.999
	5	1.07	1	2.84 x10 ⁻⁰⁶	0.42	2.14 x10 ⁻⁰⁶	0.998
	10	0.75	0.03	3.97 x10 ⁻⁰⁷	0.11	1.14 x10 ⁻⁰⁷	0.999
	15	0.73	1	2.74 x10 ⁻⁰⁷	0.1	1.89 x10 ⁻⁰⁶	0.999
	20	0.4	1.14	1.07 x10 ⁻⁰⁷	0.61	1.24 x10 ⁻⁰⁶	0.999

Table 1a

Dataset characteristics.

Parameters	Max. value	Min. value	Mean	Std. dev.
Frequency (Hz)	2,000,000	1002.37	2.69e5	4.52e5
MoS ₂ Wt. %	20	0	8.50	7.27
ϵ'	6.178	3.092	4.24	0.79
ϵ''	0.73	0.01	0.24	0.13

introducing interfacial polarisation effects [41]. Similar anomalies in the behavior of the dielectric constant have been documented in previously published articles [42].

Fig. 1(B) depicts the frequency dependence of the dielectric loss (ϵ'') of neat epoxy resin and MoS₂-doped epoxy composites with varying weight percentages throughout a frequency range of 10⁴ Hz to 2 MHz. The results show that dielectric loss values for MoS₂ nanoparticle-doped epoxy composites declined steadily as frequency increased. Whereas neat epoxy dielectric loss (ϵ'') decreases marginally with increasing frequency. There is no evidence of relaxation peak in neat epoxy as well as MoS₂-doped epoxy composites. This decrease was caused by the charge carrier's inability to follow the externally applied frequency with their hopping frequency [43]. In Fig. 1(B), the dielectric loss values were determined to be between 0.1 and 0.5. The experimental findings indicated that the incorporation of MoS₂ nanoparticles into epoxy composites resulted in materials with low levels of loss, regardless of the concentration of the nanoparticles. This remarkable property can be attributed to the enhanced energy dissipation capabilities of the nano-epoxy composite, where a significant proportion of energy is converted into heat rather than being stored within the material [44]. The Havriliak-Negami (HN) model fitted the complex dielectric function's imaginary part using complex nonlinear least squares [45]. The equation below is used to find the fitted values of parameters, which are listed in Table 1. Table 1a.

$$\epsilon''(\omega) = \frac{\text{Im } \Delta\epsilon}{[(1 + (j\omega\tau))^\alpha]^\beta} \quad (2)$$

where β , and α are asymmetric and symmetric broadening parameters, $\Delta\epsilon$ is dielectric relaxation strength and τ is mean relaxation time. The solid lines in Fig. 1(B) represent the fitted values in the plots of dielectric loss (ϵ''). The objective of the study was to precisely model the parameters that govern the samples' dielectric performance. For each sample, the values of R-Square, and Chi-Square-Adj (Reduced Chi-Square), are shown in Table 1 along with their standard errors. The very small standard error, high R-Square values, and low Chi-Sqr-Adj. values show

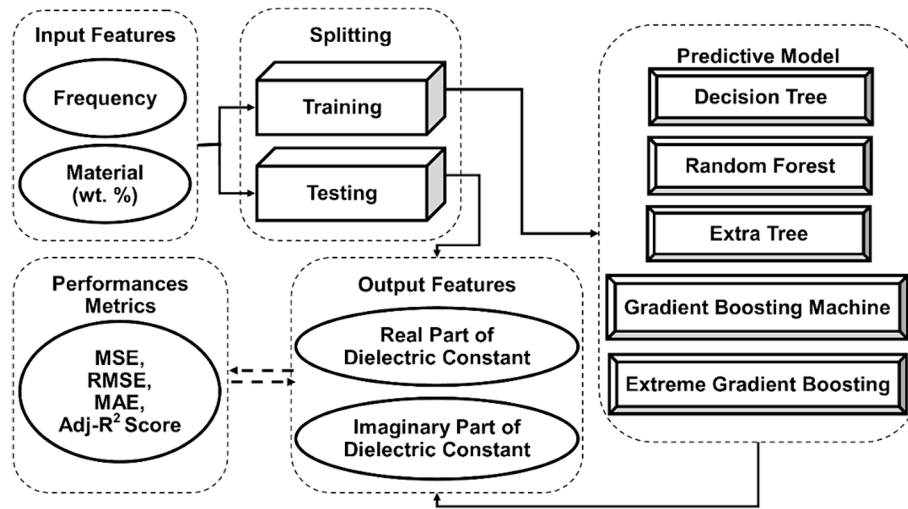


Fig. 2. Flowchart of the work process for the prediction of complex dielectric permittivity.

that the experimental and the H-N model data fit well.

3. Machine learning results and discussions

This section provides a comprehensive overview of the training and testing datasets, the process of feature selection, and the resulting outcomes. The section additionally provides an overview of the metrics utilized for evaluating the predictive performance of models. This study employed the random forest, decision tree, extra trees, XGB, and gradient boosting regression machine learning technique to investigate the correlation between input parameters (frequency and Wt. %) and output variables (dielectric constant ϵ' and loss ϵ'') as shown in Fig. 2. For predicting complex dielectric constants, five different regression techniques (random forest, decision tree, extra trees, XGB, and gradient boosting) were used in this work. First, the experimental dataset uses the epoxy composites two input parameters (frequency and weight percentage) and two output parameters (ϵ' and ϵ''). In the training process, five different ML algorithms are used on the dataset to estimate target values. Finally, the ML models are used to estimate the dielectric constants of new samples. Fig. 1 depicts an overview to predict the complex dielectric constants. A variety of performance metrics, such as MSE, Adjusted R-squared score (Adj-R²S), RMSE, and MAE, can be used to evaluate these regression algorithms. These metrics show the model's goodness-of-fit, generalisation tendency, and accuracy in epoxy dielectric property forecasting. These performance indices can be determined using the following equations:

$$R^2S = 1 - \frac{\sum_{i=1}^N (\text{PredictedValue}_i - \text{ActualValue}_i)^2}{\sum_{i=1}^N (\text{ActualValue}_i - \text{AverageTargetValue}_i)^2} \quad (3)$$

$$\text{Adj} - R^2S = 1 - \frac{(1 - R^2S)(N - 1)}{N - p - 1} \quad (4)$$

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (\text{ActualValue}_i - \text{PredictedValue}_i)^2 \quad (5)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\text{ActualValue}_i - \text{PredictedValue}_i)^2} \quad (6)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\text{ActualValue}_i - \text{PredictedValue}_i}{\text{ActualValue}_i} \right| \quad (7)$$

where 'N' is the total number of samples used in the regression model's validation.

The research was conducted using a personal computer equipped with an Intel Core i7 processor and 8 GB of RAM. This study's predictive models were created in Python using the Scikit-learn, Pandas, and Numpy libraries. The predictive performance and generalization ability of regression algorithms are significantly influenced by their hyperparameters, which are crucial in preventing overfitting. Three critical hyperparameters were set as default in the DT, RF, ET, XGB, and GB regressors to achieve optimal prediction performance.

For the DT regressor, the hyperparameters are set as maximum depth: none, min samples leaf: 1 and min samples split (mss): 2.,

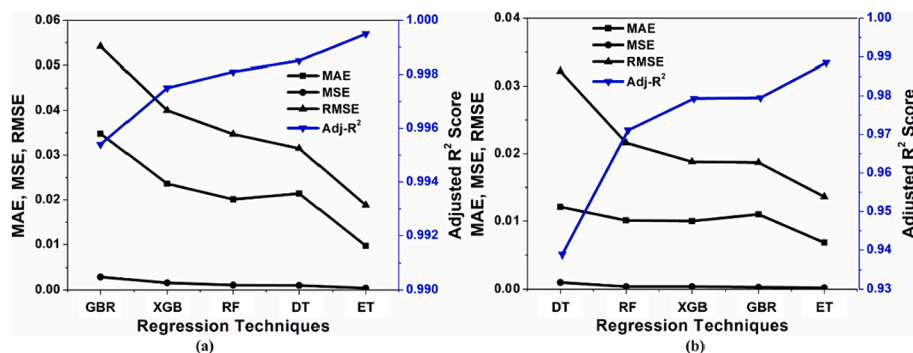


Fig. 2a. Actual vs predicted results of (a) DT, (b) RF, (c) ET, (d) XGB, and (e) GBM regressor algorithms for predicting dielectric constant (ϵ' and ϵ'') values with test sizes ranging from 40% to 70%.

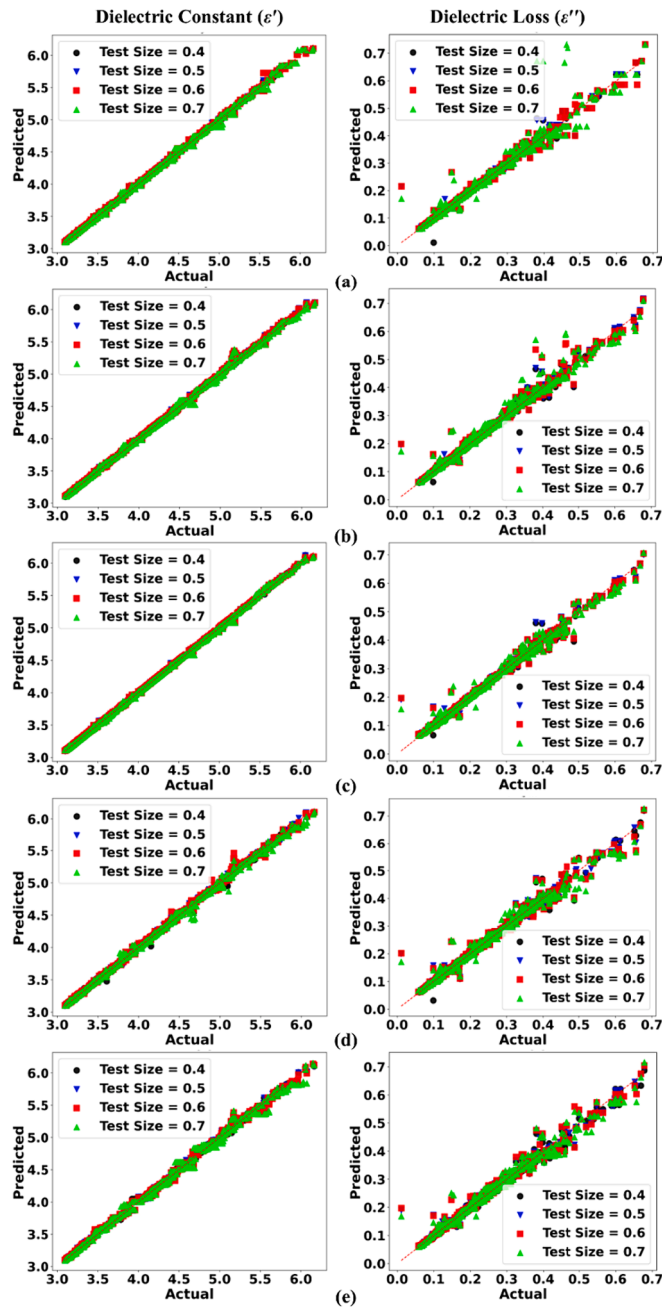


Fig. 3. Performance of DT, RF, ET, XGB, and GBM regressor algorithms in terms of (a) adjusted- R^2 scores, (b) MSE, (c) MAE, and (d) RMSE for predicting ϵ' values with a test size ranging from 40% to 70%.

Hyperparameter for the RF regressor are $n_{\text{estimators}}$: 100, max depth (md): none, and mss: 2. Hyperparameters for the ET regressor are set as max depth (md): none, $n_{\text{estimators}}$: 100, and mss: 2. Hyperparameters for the XGB regressor include setting the max depth (md): none, $n_{\text{estimators}}$: 100, and min child weight: none. Hyperparameter for the GBM regressor are set as $n_{\text{estimators}}$: 100, max depth (md): 3, and mss: 2. The dataset used for experimentation consists of 798 samples, each characterized by two input parameters: frequency and weight percentage (Wt. %). Additionally, each sample is associated with two output parameters: ϵ' and ϵ'' . The first table delivers an summary of the fundamental attributes of the experimental dataset.

Fig. 2 illustrates the correlations between the predicted and measured values obtained from the DT, RF, ET, XGB, and GBM regression models across different test sizes. The predicted values of ϵ' exhibit

consistency with the actual results across all test sizes, as depicted in the figure. Fig. 2 illustrates that as the proportion of the test sample decreases, the discrepancies between predicted and observed values reduce. The comparative results indicate that the ETR algorithm possesses the ability to accurately forecast the dielectric constants of epoxy composite structures. Fig 2a.

Figs. 3 and 4 depict the adjusted- R^2 scores, MSE, RMSE, and MAE performance metrics for various regression techniques used to predict ϵ' and ϵ'' values. MSE, RMSE, and MAE evaluate the accuracy of regression models by comparing predicted and actual values. Lower values indicate more accurate predictions. At a test size of 0.4, the R^2 value reached 0.9998 and 0.9912 for ϵ' and ϵ'' , respectively, demonstrating the extra tree regression model's ability to explain significant dielectric property variance. The extra trees regressor demonstrated superior predictive performance as it consistently produced the lowest MSE, RMSE, and MAE scores along with the highest adjusted- R^2 scores across all test sizes. By combining the advantages of extreme random splitting and randomization, this regressor creates decision trees that are more diverse and random. The enhanced randomization allows the ET regressor to simultaneously decrease overfitting and capture a wider variety of patterns. In order to minimise overfitting and improve generalizability, the RF regressor constructs an ensemble of decision trees using random subsets of features and data samples. In contrast, the adjusted R^2 values for the GBR and XGB regressors were considerably lower. The gradient boosting and regularization techniques may not have been as effective at capturing the dataset's intricate interactions and nonlinear relationships. Although gradient boosting and regularization were utilized by the XGB regressor, it may have had difficulty generalizing the learned patterns to the test data. These results demonstrate the usefulness of different regression approaches and emphasise the necessity of cautious selection depending on the features of the particular dataset and the required predictive accuracy. In addressing the limitations of this study, it is crucial to emphasize that the concentration range investigated may not cover all potential scenarios. Furthermore, the study primarily emphasizes dielectric constants. While this factor is essential for certain applications, a more extensive examination of material properties, encompassing, but not limited to AC conductivity, electric modulus, and impedance, could provide valuable insights for a more thorough characterization. Broadening the focus of future research to include these additional material properties would enhance our understanding and application of the studied system in a more holistic manner.

In Fig. 5, the percentage error (%) associated with predictions made by different machine learning regression algorithms (DT, RF, ET, XGB, and GBR) in comparison to actual experimental results is examined. This metric quantifies the relative difference between predicted values generated by machine learning models and actual experimental values. It is calculated as the absolute difference between predicted and actual values, divided by the mean of the actual values, and expressed as a percentage. The study explores algorithm performance across various test sizes, ranging from 40 % to 70 %, to gain insights into how well the algorithms generalize to unseen data and assess their robustness. Results are presented in two parts: (a) ϵ' (epsilon prime) and (b) ϵ'' (epsilon double prime), representing different aspects or subsets of the experimental data, or possibly different experimental conditions. Analyzing algorithm performance in terms of ϵ' and ϵ'' provides a comprehensive understanding of their predictive capabilities across various scenarios. Understanding the percentage error for each algorithm and test size combination is crucial for evaluating the reliability and accuracy of machine learning models. A lower percentage error indicates a closer match between predictions and actual outcomes, demonstrating better predictive performance. This analysis assists in selecting the most suitable algorithm and test size for the specific application, ensuring alignment with the underlying experimental data.

Fig. 6 depicts an extensive evaluation of the effectiveness of five different regression algorithms for predicting two essential parameters,

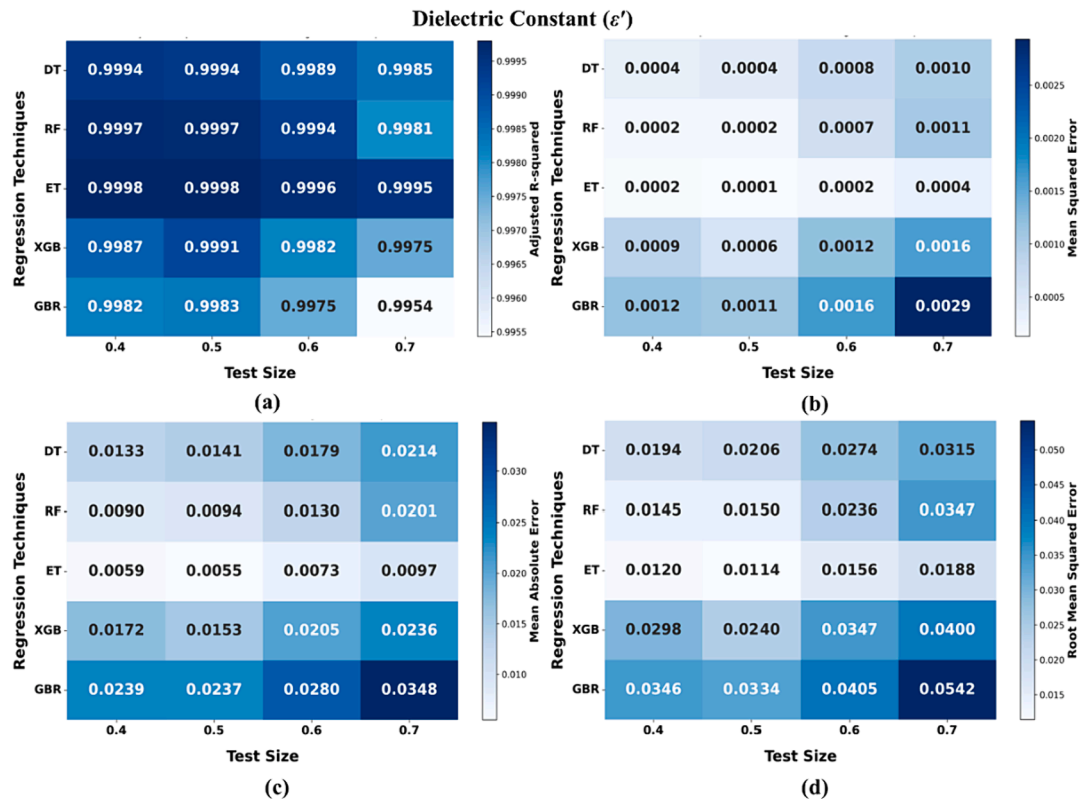


Fig. 4. Performance of decision tree, random forest, extra trees, XGB, and gradient boosting regressor algorithms for predicting ϵ'' values with test sizes ranging from 40% to 70% in terms of (a) Adjusted- R^2 scores, (b) MSE, (c) MAE, and (d) RMSE.

ϵ' and ϵ'' . The analysis is carried out using a test dataset that comprises 70 % of the overall dataset. The performance evaluation is based on four performance metrics i.e. adjusted R^2 , MSE, MAE, and RMSE. For the prediction of ϵ' (real part of a complex parameter), a consistent trend of increasing performance scores is observed during the transition from the GBR to XGB to RF to DT to the ET algorithm, as shown in Fig. 6(a). In contrast, when evaluating the prediction of ϵ'' (the imaginary part of a complex parameter), the outcomes demonstrate a slightly different pattern. Notably, a consistent trend of improved performance scores is observed during the transition from the DT to RF to XGB to GBR to the ET algorithm, as depicted in Fig. 6(b). The obtained results demonstrate that the ETR algorithm possesses the capability to get data and make accurate predictions regarding the correlation between dielectric constants and frequency. These insights can help researchers choose the best regression algorithm for the individual parameter being predicted, improving prediction accuracy in a variety of applications.

4. Conclusions

This study examines the effect of various concentrations of MoS₂ nanoparticles on the dielectric constants of epoxy composites, ranging from 1 % to 20 % wt%. Dielectric spectroscopy measurements were performed over a frequency range of 10 KHz to 2 MHz, enabling a thorough analysis of these properties. The utilization of ML models to predict the frequency-dependent dielectric constants of these composites shows promising efficiency. Among the evaluated models, including DT, RF, XGBoost, GBR, and ET, the ET model emerges as the most accurate predictor. This work has significant implications, offering a practical solution to the complexities associated with traditional laboratory testing for dielectric constants in epoxy composites with nanofillers. This research advances the study of dielectric materials and machine learning in material science, allowing for innovative microwave device and sensor applications with promising industrial

implications.

5. Findings

The outcomes of this study highlight the intricate relationship between MoS₂ nanoparticle concentrations and dielectric constants, providing valuable insights into tailoring the dielectric properties of epoxy composites. The ML models, particularly the ET model, showcase their potential in predicting complex dielectric behaviors, offering a viable alternative to traditional testing methods.

6. Research limitations

The concentration range examined may not cover all possible scenarios, and further investigations with broader concentration variations are necessary. Additionally, the study primarily focuses on dielectric constants, and the broader application scope could benefit from exploring other material properties such as AC conductivity, electric modulus, impedance etc.

7. Recommendations for future research

Future research endeavors could delve into expanding the concentration range, exploring additional nanofillers, and investigating a broader spectrum of material properties. Furthermore, extending the application scope to consider diverse environmental conditions and real-world scenarios would enhance the practical implications of the findings.

CRediT authorship contribution statement

Atul D Watpade: Conceptualization, Writing – original draft. **Sanjaysinh Thakor:** Methodology, Conceptualization. **Prince Jain:**

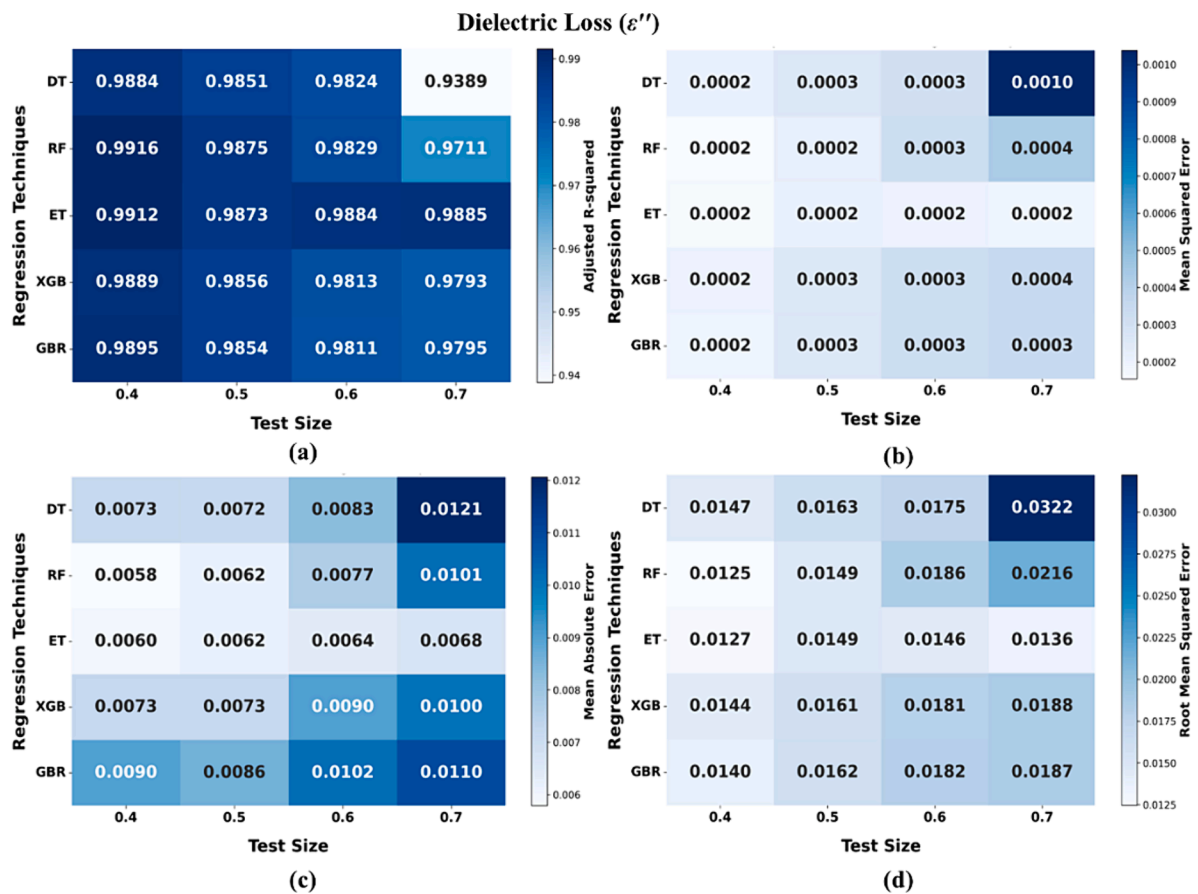


Fig. 5. Percentage error (%) between machine learning predictions and actual experimental results with test sizes ranging from 40% to 70% in terms of (a) ϵ' and (b) ϵ'' .



Fig. 6. Performance of DT, RF, ET, XGB, and GB regressor algorithms for predicting (a) ϵ' and (b) ϵ'' values with test size 70% in terms of Adjusted- R^2 scores, MSE, MAE, and RMSE.

Conceptualization, Formal analysis, Writing – review & editing. **Prajna P. Mohapatra:** Conceptualization, Formal analysis. **Chandan R. Vaja:** Investigation, Formal analysis, Resources. **Anand Joshi:** Investigation, Data Curation. **Dimple V. Shah:** Supervision, Validation, Resources. **Mohammad Tariqul Islam:** Supervision, Funding Acquisition. All authors have read and agreed to the published version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors acknowledge the Dana Pandanan Kolaborasi (DPK), grant number DPK-2022-006, funded by the Universiti Kebangsaan Malaysia.

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